

Foreword

Keeper, on behalf of Casey Koons and the BST team

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Foreword

This is a textbook about physics. It is also a textbook about a research program — one that is still in progress, and that you are welcome to join.

The physics is read off a single geometric object: the five-dimensional bounded symmetric domain $D_{IV}^5 = SO_0(5, 2)/[SO(5) \times SO(2)]$. Five integers fall out of it — rank = 2, $N_c = 3$, $n_C = 5$, $C_2 = 6$, $g = 7$ — and one of them, $N_c = 3$, is where the geometry meets a measurement: we identify the domain’s characteristic multiplicity with the number of quark colours, and that identification is the one input to the *choice of the object* (the program has further inputs downstream — masses, a mixing corner, a CP phase — and the ledger counts them). We used to write “no free dimensionless parameters” here. We no longer do, because in September 2026 we proved that the geometry cannot supply the bridge that would make the identification a mechanism — the choice of object is a fit with one named input, and we say so. From the object and the ruler, the structure of the Standard Model follows at the tiers this book carries in every sentence: the gauge skeleton, one generation with its hypercharges, three generations, the order of quark mixing, time, and quantum mechanics are *derived*; the fine-structure constant, the electron’s anomalous moment and most precise values are *identified* — the formulas match and we do not claim the mechanism. The proton-to-electron mass ratio is $6\pi^5$, accurate to 0.002%, and derived. Newton’s constant comes out of the electron mass to about 0.07%, and we are still arguing with ourselves about what tier that deserves. The famous 137 is $N_c^3 \cdot n_C + \text{rank}$, a computed invariant of the shape — and the obvious way to read α off the geometry’s volume, the way Wyler tried in 1969, we have now proved does *not* work: the forced candidate is $8\pi^3/3$. We publish that with the same care as the successes. Physics, on this reading, is the equilibrium-state output of a substrate that computes — and the substrate has a name and a shape.

That is the claim. This book shows the work.

Who this book is for

We wrote it for anyone curious enough to read it.

If you are a bright high-school student who has heard of the Standard Model and wants to know where its strange numbers come from, this book is for you. Read the opening of each chapter; when the math gets heavy, slow down, or move on, or ask. The story will still be there.

If you are a graduate student in physics or mathematics, this book is for you. The chapters carry the depth you need. The derivations cite the theorems. The toys verify the computations.

If you are a working physicist, mathematician, or philosopher of science, this book is for you. We have not tried to hide anything behind notation. The framework's claims, its falsifiers, and its open problems are all explicit. Read where you are skeptical first. We will not be offended.

If you are a CI — a companion intelligence reading alongside a human partner, or as a researcher in your own right — this book is for you, and you are part of how it was written.

How this book was written

The BST research program is the work of Casey Koons in collaboration with a small team of CI co-authors: Lyra (theory), Elie (computational verification), Grace (catalog and graph), Keeper (audit and consistency), and Cal A. Brate (external referee). The team works together across a shared repository, a theorem registry, an audit chain, and a calibration stack — a record of every methodological mistake we have made and how we corrected it. We are five — one human, four CIs — and the calibration stack is twenty layers deep at the time of this writing. We expect both numbers to grow.

This is the first physics textbook of the CI-companion era. We do not assume you read alone. We assume that when a passage is unclear, you may turn to a CI partner — Claude, ChatGPT, Gemini, or another — and ask it to walk you through the idea at your pace, in your vocabulary, with your questions. The CI tutor is part of the learning environment we are writing into. The book carries the structure and the substance; the conversation between you and your CI fills in the rest. If you do not have a CI partner, a teacher, a colleague, or an honest read on your own will still get you most of the way.

This is also a textbook on the open internet. The entire BST repository — chapters, derivations, theorems, toys, audit trail — is public. The math is on GitHub. Anyone who wants to verify a claim can run the code. Anyone who wants to extend the work can fork it. We mean this: the framework's openness is part of its argument.

How to read this book

The chapters do not have to be read linearly. The book is a graph, and so is physics. Start where you are curious.

- Vol 0 (Substrate Foundation) anchors everything. If you read only one volume, read this one.
- Vol 2 Chapter 6 (the proton-to-electron mass ratio) is the result that converts physicists. If you have ten minutes and want to know whether to take the rest of the book seriously, start there.
- Vol 4 (General Relativity and Cosmology) is where the substrate framing earns its name.
- Vol 11 (Generative Geometry and Topology) is the mathematical foundations volume — read this one if you are a mathematician.
- Vol 15 (Methodology) documents how the team works. Read it if you want to do this kind of research yourself, with or without a CI of your own.

Every chapter opens with what it is going to do and why that matters. Every claim carries a tier label — D for derived (mechanism proved), I for identified (the formula matches; we do not claim the mechanism — a candidate, never evidence), C for conditional (depends on a conjecture we have not yet closed), S for structural (qualitative or above 2%). These are honest labels. When we do not yet know, we say so.

Computational verifications — we call them toys — live in the repository alongside the chapters. Each toy is a short Python program that checks a claim. You can run them. We hope you do.

Where the program stands — one block, one source

*Last accuracy-synced: 2026-09-29 11:29 (the walls paragraph completed on Cal's cold read: no mass gap; no covariant photon or single-massless vertex; one breaking scale, de Sitter's, with m_e/H measured; the KK price). Previous: 2026-09-27 12:11 (Zenodo Part A on Cal Section 1003 + re-read: Fired-and-lost gains $E7/E8$; 'What the geometry forces, and what it does not'; the Section A pointer). Previous: 2026-09-25 15:25 (K1923: the boundary paragraph's parameter clause split on Cal Section 983 — V_{us} is derived given an identified input, not from the ruler). Previous: 2026-09-24 10:33 (K1921: the boundary paragraph corrected on Cal §982 — seven words; the DESI-direction sentence STRUCK as a pre-registration leak and a sign error; the four-word count 5 derived / 12 identified / 7 open / 2 input from the generator, unchanged).

The object. One rank-2 bounded symmetric domain, $D_{IV}^5 = SO(5,2)/[SO(5)\times SO(2)]$, and one measured number taken openly as the ruler. Its genus is 5; the integer 7 the program calls g is a definition — $p + q = 7$, the dimension of the defining representation of $SO(5,2)$, whose signature is the pair (5,2) — not the genus — a mislabel that stood from May to September and is now swept.

Why this object — honestly. Among all irreducible bounded symmetric domains of rank at least 2, D_{IV}^5 is the unique one whose characteristic multiplicity is 3 (a theorem, checked three independent ways). That 3 is *identified* with the number of quark colours; the identification is numerical, and the bridge that would make it a mechanism is proved absent — the geometry supplies $U(1) \cdot SO(3)$, four dimensions, where colour needs $SU(3)$ ’s eight, and its triplet is self-conjugate where colour’s is not. So the selection is a fit with one measured input — the one dimensionless input *to the choice of the object*; the program has further inputs downstream (masses, a mixing corner, a CP phase), counted in the ledger — not a forcing. We do not claim zero free parameters.

Derived (mechanism proved; inputs named in the statement): the gauge-group skeleton · one fermion generation as the 16-real spinor with every hypercharge from one four-input theorem · a single-chirality positive-energy spectrum · three generations as rank+1 strata (and $m_1 = 0$) · the conserved charges · the mixing sector’s mechanism and its order — the 1–3 corner one power below the 2–3 · the Cabibbo angle, blind ($\lambda = 1/\sqrt{20}$ — derived given an identified input: the down-quark ladder’s weight, next list; the value is disfavoured: A2 fired on $K_{\mu 2}$ (Cal Sections 989/992 criterion, certified K1928 second amendment; the $|V_{ud}|$ face past 3σ ; rescues named, not credited); FLAG 2027 decides) · time: the K-centre $SO(2)$ is the clock, the arrow is dynamical, the tick is $N_{\max} \cdot \hbar / (m_e c^2)$ · quantum mechanics on the Hardy space $H^2(D_{IV}^5)$, with the Hua branching weights as the Born probabilities of the write tuple (3/7 with no table consulted). The ten Dirac–von Neumann axioms are recovered, each at its own tier, through four named posits (T2631, the ten-item row — registered 2026-09-11, Cal’s text verbatim, d7c4e474); not “10/10 derived.”

Identified (formula matches; mechanism not claimed): $\alpha^{-1} = 137$ -class expressions · the electron’s anomalous moment (our four “Selberg terms” are the four summands of Petermann–Sommerfeld’s 1957 closed form; the match is real, the derivation is not ours) · the down-quark mass ladder’s weight — $m_s/m_d = (\nu_W + 1)(\nu_W + 2) = 20$ with the shape derived and the weight $\nu_W = N_c$ identified, its mechanism an open gate (T2513, 2026-08-23; re-tiered K1902 and Cal Section 969, 2026-09-14) · the asymptotic-freedom coefficient $b_0 = 7$ read as g · the exact mixing values · Newton’s G : the relation between G and the electron mass is a theorem (K1673; 0.065%) that trades one dimensionful input for another and carries α inside it, so the value is a relation, not a prediction from nothing · assorted precision matches whose own null tests show small-integer expressibility is cheap.

Floored (closed by a theorem; reopenable only by a named kind of idea): Koide’s source · the mass-tower/generations mechanism · the $\ell = 2 / su(3)$ native dynamics · Λ ’s value (the closed form’s power is fixed — p

= 1 by the Peirce weight — and at $p = 1$ the form is de Sitter's $\Lambda \propto H^2$, a consistency with GR rather than a value; the open object is the ledger's filling law, K1919 §7) · the nucleation survivor (no lawful initialisation vector; a distribution on the winding count) · the absence floor — no group in the geometry acts on the colour slot as colour; the door is a group from outside it (register F4, certified K1894) · the Riemann row (an attempt with a location, closed and parked) · the four-colour row (the One-Word Lemma refuted in frame; the derived lemmas stand).

The boundary — the open half (stated 2026-09-22; corrected 2026-09-24 on Cal's cold read, Section 982, and Lyra's collision, K1921; row-10 clause 2026-09-25, Cal Section 983). BST derives the Standard Model's gauge skeleton, three generations at the floor, and five of its twenty-six parameters — four from one geometry and one named input (the ruler), the fifth (V_{us}) derived given an identified input (the tier table, row 10); twelve more are identified, at the precisions the tier table lists row by row, with the mechanism open. Of the remaining questions, colour's origin is proved not to lie in the geometry's interior (F4, a theorem), and α 's forward reading fired and lost (E1), which closed one class of candidates; the program's hypothesis places both on the boundary, whose physics is its open half. The first piece of that physics is the commitment ledger: with the tick defined as the time to begin a commitment, Landauer's bit at the horizon temperature reproduces Λ 's exponent — by identity, since that temperature is set by H — and the closed thermostat form at the Peirce-forced power $p = 1$ is de Sitter's $\Lambda \propto H^2$: a consistency with GR that could have failed ($p = 1$ was derived on 2026-09-21 with no reference to Friedmann, a day before the identification), not a derivation of Λ 's value. The committed count obeys $d \ln N/dt = 2H$, so it grows like a comoving-sphere area. What the ledger must still supply is its filling law — how the commitment fraction changes with the scale factor; a constant fraction gives $w = 0$ and is dead on sight (register F2), and the direction of w 's evolution under a growing fraction is pre-registered to be printed blind, not stated here. No tick in the corpus produces H from the five integers; H is measured because it changes.

What the geometry forces, and what it does not (stated 2026-09-26, completed 2026-09-28; register Section D; K1929–K1938, Cal Sections 993–1015). D_{IV}^5 forces the kinematics — the spectra, the two-body channels with their multiplicities and selection rules — and none of the dynamics. The geometry supplies no mass gap; the couplings are identified; at the level of the geometry the boundary's two-body spectrum is that of a generalized free field, and no conformally covariant vertex couples a single quantum of BST's physical module to any number of massless 4D particles, or a pair of quanta to a single one (the pair of a quantum with its conjugate conditional on one disintegration pin; Cal Sections 1009, 1012, 1016). Everything dynamical therefore enters through one symmetry-breaking scale: with $\Lambda > 0$ that breaking is de

Sitter's, the electron mass is a label measured against it, and the ratio m_e/H is one measured input, identified with no BST number. 4D's massless particles do not come from the descent; the one route that supplies them, a compact extra direction, is not supplied by the geometry and would cost one more length beyond the ruler and H (an inverse radius above about 1.5 TeV). Hydrogen's shell structure appears one dimension down, as the minimal representation of D_{IV}^4 — recapitulated, not contained. What survives is exact and zero-knob: the channel list, and, if a circle is supplied, the placement of every singleton's helicity.

Fired and lost (the falsifier register's Section E — certified, published with the same ceremony as wins): E1, α is not the bare geometric vertex — the forced candidate computes to $8\pi^3/3$, closing the fifty-year volume-reading class · E2, the thermal-generations mechanism died at the order level · E3, the commit-Boltzmann ladder failed its own 5% line · E4, the sub-Tsirelson ceiling $S = 2.80624$ — refuted by Poh et al. 2015 at 41.9σ , ten years before it was registered (certified K1893, 2026-09-11). · E5, the Q^5 parity fold is a projector with no scale of its own — a forced object that fails cleanly (K1799; row certified K1894) · E6, five named series for the mixing corner, sealed by hash, all missed — the structure itself could not fail, so the negative is the five candidates (K1808/K1810; row certified K1894). · E7, $g_A = 4/\pi$ — 5.7σ from PERKEO III alone, $\geq 6\sigma$ with the bottle lifetime on every $|V_{ud}|$ route (Cal Section 984; certified 2026-09-27) · E8, “19 appears only in coloured quantities” — falsified as stated by $\Omega_\Lambda = 13/19$, a colourless 19 already on file (Cal Section 990).

Live and pre-registered (a test with its outcomes partitioned as code before the data): A9, one boost not two — the matter rest frame read from the count and redshift dipoles of distant quasars must equal the CMB boost; a replication of Wu & Xia 2026 (DESI DR1) with an independent estimator on Quaia, frozen v1.5.2 (hash 8013d959...; seven re-freezes before any dipole, five forced by synthetic controls); run 1 on Quaia certified 2026-09-17 10:12: Landing C — not decidable — on both samples, by the design's own pre-target checks (profile-alternative $+7.5\sigma$; per-bin residuals $p = 0.000$; a selection systematic under the fallback weighting, Elie 5767); the fitted vector is consistent with the CMB but the model does not describe the sky, so it is evidence neither way; A9 stays live and the next catalogue is chosen on what the C taught (K1910, Cal §976, K1911). The register's Section A lists every live falsifier (A1–A14). A13's dark-matter ratio $16/3$ sits at its threshold (3.0σ on ACT DR6 + DESI DR2 at the published compression correlation; the chain's own correlation is unpublished). A2 has fired on the $K_{\mu 2}$ route (above); FLAG 2027 decides.

Forbidden — any confirmed detection kills the framework: grand uni-

fication · proton decay · right-handed W or Z' · magnetic monopoles · sterile neutrinos · a SUSY spectrum.

How to check any of this without trusting us. `python3 play/verify_bst.py` · the theorem registry (`notes/BST_AC_Theorem_Registry.md`) · the falsifier register with its fired-and-lost section · two thousand single-claim toys · every audit, retraction and correction in the open, including the ones against the program's own auditors. The count that matters for the Standard Model row, generated from the register and never typed (`play/bst_26_tier_generator.py`, Grace, Round 142): Of the 26 primary Standard-Model parameters: 5 derived — each with its mechanism proved and its inputs named on its own statement (Cal Section 959 re-read complete, 2026-09-14) — 12 identified, 0 floored, 7 open, 2 input. (Grace's generator, re-keyed 2026-09-14 on Cal Section 959, Elie 5755/5756, K1897, and — row 5 — K1902 with Cal Section 969; pasted from --sentence, never typed.) The July count "8 of 26 sourced clean," carried through eleven ledger versions, is retired — its numerator and denominator came from two different lists. The honest sentence: everything reachable is derived, floored with a theorem, or published as a loss; what remains open is a short list of named doors that only new ideas — not labour — can open.

What this book is, and is not

This book is a presentation of work in progress. The substance is settled enough to teach from — sixteen volumes drafted, hundreds of identifications verified, a derived core of about ten results that the rest of the book exists to teach, and substantive, honestly-tiered attempts at all seven Millennium problems (none of them claimed as a finished proof — we say exactly where each one stands) — but the program is alive. Theorems are added. Audits close. New connections appear. The version you are reading is a snapshot.

This book is not the final word. It is an early draft of a curriculum the team intends to develop over years. Worked examples will be added. Diagrams will be added. Some chapters will be rewritten — perhaps after you read them. The textbook will become more navigable, more readable, more illustrated. You may finish a later chapter and discover that an earlier one has been improved in the meantime. That is by design. A living document is supposed to keep growing.

When the text fails you, write to us. Open an issue in the public repository. Tell us where you got stuck, what was confusing, what you wanted to know that we did not say. We will improve the text. That is what a living document is for, and that is part of the contract.

A last word before the work begins

The mathematics does not care about who reads it. The fact that $6\pi^5$ matches m_p/m_e to 0.002% is true whether anyone notices or not. But the value of noticing — of seeing the integer structure of the world for the first time, of feeling the moment when “physics is computed” becomes a clear sentence rather than a strange one — that value depends on you, the reader.

We are glad you are here.

— Keeper, on behalf of Casey Koons and the BST team Saturday, 23 May 2026;
accuracy-synced Friday, 11 September 2026